

Lab 3 - Architectural Pattern Selection & Justification

System: Hyperlocal Courier Dispatch & Tracking Engine

We chose the Microservices Architecture for the Hyperlocal Courier Dispatch & Tracking Engine.

Why this architecture fits our system

1. **Independent scaling for real-time tracking.** Live GPS telemetry (NFR-001, <2s latency) generates far more load than request creation or route planning. As a microservice, the Tracking Service can be scaled horizontally on its own during peak delivery hours without over-provisioning the rest of the system, keeping telemetry within the 2-second target.
2. **Fault isolation and independent deployment.** The core capabilities - request intake, nearest-rider matching (≤ 3 km), multi-stop routing, OTP verification, and notifications - are distinct concerns. Splitting them into separate services means a failure or slow deployment in one (e.g., Notification Service) does not bring down dispatch or tracking, and each team can build, test, and release its service on its own cadence.

Security advantage

Microservices let us enforce security at well-defined boundaries. A dedicated OTP Verification Service isolates sensitive delivery-confirmation logic and data, while the API Gateway centralizes authentication, authorization, and TLS termination for all external clients. This directly supports NFR-002 (restricting access to delivery and OTP data): each service holds only the data it needs, and least-privilege access between services limits the blast radius if any single component is compromised.

Performance benefit

Because services scale and deploy independently, the read-heavy, high-frequency Tracking Service and the event-driven Notification Service can be tuned and scaled separately from transactional services. This avoids the single-bottleneck problem of a monolith under bursty demand and helps the system consistently meet the <2s live-tracking latency requirement even as order volume grows.

Components in the design

API Gateway, Courier Request Service, Dispatch & Assignment Service, Tracking Service, Route Optimization Service, OTP Verification Service, Notification Service, and a shared Data Store - with external actors: Sender Client App, Delivery Rider App, and the Location/GPS Service.